Model-Implied Instrumental-Variable and James–Stein Estimation of Structural Equation Models: A Case Study With lavaan

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Abstract

This article sets out the notation of a linear structural equation model with a mean structure and describes model-implied instrumental-variable two-stage least squares (MIIV-2SLS) and the scaling and aggregated James–Stein estimators as implemented in lavaan. Both obtain factor loadings and latent regression slopes equation by equation without iterative optimization. The remaining variance and covariance parameters are subsequently estimated by reweighted least squares. For both procedures we state the estimators of the factor loadings and latent regression slopes, the subsequent estimation of the variance, covariance, and mean structure parameters, and the default covariance estimators of the factor loadings and latent regression slopes. For the James–Stein estimators we write the complete estimation procedure as a map from the sample moments to the parameter estimates and give the resulting delta method standard errors.

Keywords: structural equation modeling, model-implied instrumental variables, James–Stein estimation, two-stage least squares, delta method

Structural equation models (SEMs) with latent variables are usually estimated by maximum likelihood (ML), which estimates all parameters jointly. Despite its desirable properties when its assumptions hold, ML has limitations that can arise in practice. For instance, a misspecification in one equation can bias estimates in other equations (Bollen et al., 2007; Bollen, 2019; Dhaene & Rosseel, 2023; Kaplan, 1988; see also Jin et al., 2021, for diagonally weighted least squares), and the iterative optimization can fail to converge or end in inadmissible solutions, especially in small samples (Bollen, 2019; Dhaene & Rosseel, 2023). Equation-by-equation estimators offer an alternative (e.g., Bollen, 1996; Burghgraeve et al., 2021). In the approaches considered, factor loadings and latent regression slopes are estimated one equation at a time and in closed form, which may limit the propagation of misspecification across equations and avoids iterative optimization for this part of the estimation. Two such estimators are considered. Model-implied

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instrumental-variable (MIIV) estimation replaces each latent variable by a scaling indicator and uses instruments implied by the model to correct for the resulting measurement error (Bollen, 1996, 2019; Bollen & Bauer, 2004). James–Stein estimation, proposed for SEMs by Burghgraeve et al. (2021), instead replaces each latent predictor by an estimate of its conditional expectation given observed proxies and then applies least squares. This conditional expectation is estimated using James–Stein type shrinkage (Efron & Morris, 1973, 1975; James & Stein, 1961). The approach follows the principle of regression calibration in errors-in-variables regression (Carroll et al., 2006; Fuller, 1987; Whittemore, 1989). Moreover, its aggregated version uses reliability weighted sums of indicators as proxies and was found to outperform ML and MIIV in terms of mean squared error under the small sample conditions examined (Burghgraeve et al., 2021). Both versions were further evaluated for confirmatory factor models by Dhaene and Rosseel (2024).

Beyond point estimation, the usefulness of these approaches for statistical inference depends on the accuracy of their standard errors. For MIIV, conventional standard errors for factor loadings and latent regression slopes follow from two-stage least squares (Bollen, 1996). For variance and covariance parameters estimated subsequently by unweighted least squares, standard errors accounting for the previously estimated loadings and latent regression slopes were derived by Bollen and Maydeu-Olivares (2007) and generalized to other weight matrices by Jin et al. (2021). These calculations require derivatives of the estimated loadings and latent regression slopes with respect to the sample moments, for which Fisher and Bollen (2020) provided explicit analytical expressions. More recently, Maydeu-Olivares et al. (2026) compared subsequent variance and covariance estimation procedures and the accuracy of their standard errors. For James–Stein estimation, Burghgraeve et al. (2021) derived a variance expression for latent regression estimates that accounts for the estimated reliability adjustment. However, their empirical evaluations used bootstrap standard errors for both the scaling and aggregated estimators, and they did not provide a complete analytical standard error calculation covering all estimation stages. Both approaches are now implemented in the R package lavaan (Rosseel, 2012), where analytical standard errors for the scaling and aggregated James–Stein estimators are obtained by applying the delta method to the complete estimation procedure (Rosseel, 2026a, 2026b).

Present Study and Contributions

This article makes three minor contributions. First, it describes and conceptually compares MIIV and James–Stein estimation within a common notation, following their implementation in lavaan. Second, it develops and illustrates a complete delta method calculation of standard errors for the scaling and aggregated James–Stein estimators, accounting for uncertainty across all estimation stages. Third, it evaluates their standard error accuracy in a simulation study comparing both James–Stein estimators with MIIV and ML.

The remainder of the article introduces the model and notation, describes the estimation of factor loadings, latent regression slopes, and the remaining parameters, presents the analytical standard errors, and reports the simulation study.

Model, Notation, and Parameterization

Following the all-y LISREL notation (Jöreskog & Sörbom, 1996, submodel 3B, Chapter 6; Rosseel & Vidal, 2025), let $\mathbf{y} \in \mathbb{R}^p$ contain the continuous observed variables and $\boldsymbol{\eta} \in \mathbb{R}^q$ the latent variables. The measurement and latent variable models are

$$\mathbf{y} = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\eta} = \boldsymbol{\alpha} + \mathbf{B}\boldsymbol{\eta} + \boldsymbol{\zeta},$$

where $\boldsymbol{\nu} \in \mathbb{R}^p$ contains the measurement intercepts, $\boldsymbol{\Lambda} \in \mathbb{R}^{p\times q}$ contains the factor loadings, $\boldsymbol{\alpha} \in \mathbb{R}^q$ contains the structural intercepts, and $\mathbf{B} \in \mathbb{R}^{q\times q}$ contains the latent regression slopes, that is, the regression slopes among the latent variables. The measurement errors $\boldsymbol{\epsilon} \in \mathbb{R}^p$ and structural disturbances $\boldsymbol{\zeta} \in \mathbb{R}^q$ have zero means. Let $\mathbb{S}_+^k$ and $\mathbb{S}_{++}^k$ denote the sets of symmetric positive semidefinite and positive definite $k\times k$ matrices, respectively. Assume $\mathrm{Var}(\boldsymbol{\epsilon}) = \boldsymbol{\Theta} \in \mathbb{S}_+^p$, $\mathrm{Var}(\boldsymbol{\zeta}) = \boldsymbol{\Psi} \in \mathbb{S}_+^q$, and $\mathrm{Cov}(\boldsymbol{\epsilon}, \boldsymbol{\zeta}) = \mathbf{0}_{p\times q}$.

Provided that $\mathbf{I}_q - \mathbf{B}$ is nonsingular, the latent and observed mean vectors are

$$\boldsymbol{\mu}_\eta = (\mathbf{I}_q - \mathbf{B})^{-1}\boldsymbol{\alpha} \in \mathbb{R}^q, \qquad \boldsymbol{\mu}_y = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\mu}_\eta \in \mathbb{R}^p,$$

where $\mathbf{I}_q$ is the $q \times q$ identity matrix. The implied observed covariance matrix is

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}(\mathbf{I}_q - \mathbf{B})^{-1}\boldsymbol{\Psi}(\mathbf{I}_q - \mathbf{B})^{-\top}\boldsymbol{\Lambda}^\top + \boldsymbol{\Theta}.$$

Both estimation procedures considered here require a scaling (marker) indicator parameterization (Bollen, 1996, 2019; Burghgraeve et al., 2021), in which one loading per factor is fixed to one and the corresponding measurement intercept is fixed to zero (Bollen, 1989). Hence, let $\mathbf{m} \in \mathbb{R}^q$ collect the scaling indicators from $\mathbf{y}$, ordered so that the k-th element is the scaling indicator of η_k and loads on η_k only, and let $\boldsymbol{\epsilon}^{(m)} \in \mathbb{R}^q$ collect the corresponding measurement errors. Then $\mathbf{m} = \boldsymbol{\eta} + \boldsymbol{\epsilon}^{(m)}$ and $E(\mathbf{m}) = \boldsymbol{\mu}_\eta$.

Let $\boldsymbol{\theta} \in \mathcal{T} \subseteq \mathbb{R}^d$ collect the free parameters, where $\mathcal{T}$ incorporates identification restrictions. Assume that $\boldsymbol{\Sigma}(\boldsymbol{\theta}) \in \mathbb{S}^p_{++}$. Denote the sample mean vector and covariance matrix by $\bar{\mathbf{y}} \in \mathbb{R}^p$ and $\mathbf{S} \in \mathbb{S}^p_+$, respectively, computed from a single group of independent and identically distributed observations $\mathbf{y}_i$, $i = 1, \ldots, n$, of $\mathbf{y}$, which is the setting of all estimation expressions below.[1] Define

$$\mathbf{s} = \begin{pmatrix} \bar{\mathbf{y}} \\ \operatorname{vech}(\mathbf{S}) \end{pmatrix} \in \mathbb{R}^h, \qquad \boldsymbol{\sigma}(\boldsymbol{\theta}) = \begin{pmatrix} \boldsymbol{\mu}_y(\boldsymbol{\theta}) \\ \operatorname{vech}\{\boldsymbol{\Sigma}(\boldsymbol{\theta})\} \end{pmatrix} \in \mathbb{R}^h, \qquad h = p + \frac{p(p+1)}{2},$$

where vech stacks the lower triangular entries, including the diagonal.

Estimation of Factor Loadings and Latent Regression Slopes

Model-implied instrumental-variable estimation and James–Stein estimation obtain factor loadings and latent regression slopes equation by equation (Bollen, 1996, 2019; Burghgraeve et al., 2021). Both are estimated from centered observations, with intercepts estimated subsequently. The following expressions assume that the loadings and latent regression slopes being estimated are free and not subject to equality constraints.

Model-Implied Instrumental-Variable Estimation

In model-implied instrumental-variable (MIIV) estimation (Bollen, 1996, 2019), the scaling indicator equations are rearranged as

$$\boldsymbol{\eta}_i = \mathbf{m}_i - \boldsymbol{\epsilon}_i^{(m)}$$

and substituted into the remaining measurement and latent variable equations. This latent to observed variable (L2O) transformation produces equations involving observed dependent and explanatory variables, with composite errors containing the original errors or disturbances and scaling indicator measurement errors. The intercepts, factor loadings, and latent regression slopes remain unchanged.

After centering and stacking observations, the j-th L2O equation has the representation

$$\mathbf{d}_j = \mathbf{X}_j\mathbf{a}_j + \mathbf{u}_j,$$

where $\mathbf{d}_j \in \mathbb{R}^n$ contains the centered dependent variable, which is an indicator in a measurement equation or the dependent factor's scaling indicator in a latent variable equation. The matrix $\mathbf{X}_j \in \mathbb{R}^{n \times k_j}$ contains the centered scaling indicators of the k_j latent predictors, $\mathbf{a}_j \in \mathbb{R}^{k_j}$ contains the corresponding loadings or latent

[1] Following lavaan, $\mathbf{S}$ uses the divisor n for the James–Stein estimators and $n - 1$ for MIIV estimation.

regression slopes, and $\mathbf{u}_j \in \mathbb{R}^n$ contains the composite errors. The explanatory variables in $\mathbf{X}_j$ generally correlate with $\mathbf{u}_j$ through their shared scaling indicator measurement errors.

The hypothesized model identifies observed variables that are uncorrelated with the composite error and sufficiently related to the explanatory variables. Let $\mathbf{Z}_j \in \mathbb{R}^{n\times \ell_j}$ contain these centered instruments, with $\ell_j \geq k_j$. In the first stage, the explanatory variables are regressed on the instruments, giving

$$\widehat{\mathbf{X}}_j = \mathbf{P}_j\mathbf{X}_j \in \mathbb{R}^{n\times k_j}, \qquad \mathbf{P}_j = \mathbf{Z}_j(\mathbf{Z}_j^\top\mathbf{Z}_j)^{-1}\mathbf{Z}_j^\top \in \mathbb{R}^{n\times n}.$$

In the second stage, the dependent variable is regressed on these fitted explanatory variables. The resulting MIIV two-stage least squares (MIIV-2SLS) estimator is

$$\widehat{\mathbf{a}}_{j,\mathrm{MIIV}} = (\widehat{\mathbf{X}}_j^\top\widehat{\mathbf{X}}_j)^{-1}\widehat{\mathbf{X}}_j^\top\mathbf{d}_j = (\mathbf{X}_j^\top\mathbf{P}_j\mathbf{X}_j)^{-1}\mathbf{X}_j^\top\mathbf{P}_j\mathbf{d}_j,$$

assuming that $\mathbf{Z}_j$ has full column rank and that $\mathbf{Z}_j^\top\mathbf{X}_j$ has rank k_j.

Scaling James–Stein Estimation

The James–Stein approach replaces the latent predictors in each equation by estimates of their conditional expectations given observed proxies, followed by least squares (Burghgraeve et al., 2021). The implementation considered here is restricted to recursive latent variable models without higher order factors. Let $\mathcal{P}_j$ index the k_j latent predictors of equation j. The proxies are indicators that load only on the corresponding latent predictor, with measurement errors $\boldsymbol{\epsilon}_j^{(p)}$ such that $\mathrm{Var}(\boldsymbol{\epsilon}_j^{(p)})$ is diagonal and $\mathrm{Cov}(\boldsymbol{\epsilon}_j^{(p)}, \epsilon_{d_j}) = \mathbf{0}$, where ϵ_{d_j} is the measurement error of the dependent variable. For the latent variable equation of η_k, additionally $\mathrm{Cov}(\zeta_k, \boldsymbol{\eta}_{\mathcal{P}_j}) = \mathbf{0}$. In the scaling version, the proxies are the scaling indicators, whose equations are

$$\mathbf{m}_{i,\mathcal{P}_j} = \boldsymbol{\eta}_{i,\mathcal{P}_j} + \boldsymbol{\epsilon}_{i,\mathcal{P}_j}^{(m)},$$

where all three vectors belong to $\mathbb{R}^{k_j}$. Define

$$\boldsymbol{\Sigma}_{mm,j} = \mathrm{Var}(\mathbf{m}_{i,\mathcal{P}_j}) \in \mathbb{S}_{++}^{k_j}, \qquad \boldsymbol{\Omega}_j = \mathrm{Var}(\boldsymbol{\epsilon}_{i,\mathcal{P}_j}^{(m)}) \in \mathbb{S}_{+}^{k_j}.$$

Under joint normality,

$$E(\boldsymbol{\eta}_{i,\mathcal{P}_j} \mid \mathbf{m}_{i,\mathcal{P}_j}) = \boldsymbol{\mu}_{\eta,\mathcal{P}_j} + \mathbf{R}_j\left(\mathbf{m}_{i,\mathcal{P}_j} - \boldsymbol{\mu}_{\eta,\mathcal{P}_j}\right),$$

where $\boldsymbol{\mu}_{\eta,\mathcal{P}_j} \in \mathbb{R}^{k_j}$ and $\mathbf{R}_j = (\boldsymbol{\Sigma}_{mm,j} - \boldsymbol{\Omega}_j)\boldsymbol{\Sigma}_{mm,j}^{-1} \in \mathbb{R}^{k_j\times k_j}$.

Note that, without normality, the same expression gives the best linear predictor under the stated covariance assumptions (Burghgraeve et al., 2021). Preliminary measurement error variances are obtained using a Spearman type procedure (Burghgraeve et al., 2021; Rosseel, 2026b). For indicator e with sample variance s_{ee}, the preliminary error variance is $s_{ee}(1-\widehat{h}_e^2)$, where the communality $\widehat{h}_e^2$ averages $s_{ef}s_{ef'}/(s_{ff'}s_{ee})$ over all pairs (f, f') of other indicators of the same latent variable. Incorporating the small sample correction gives

$$\widehat{\mathbf{R}}_j = \left(\mathbf{S}_{mm,j} - c_n\widehat{\boldsymbol{\Omega}}_j\right)\mathbf{S}_{mm,j}^{-1}, \qquad c_n = \frac{n-3}{n-1},$$

where $n > 3$, $\mathbf{S}_{mm,j} \in \mathbb{S}_{++}^{k_j}$ is the sample covariance matrix of the scaling indicators, and $\widehat{\boldsymbol{\Omega}}_j \in \mathbb{S}_{+}^{k_j}$ is the diagonal matrix of their preliminary measurement error variance estimates. These preliminary error variance estimates are used for prediction and need not equal the final model estimates.

Using the scaling indicator sample means $\overline{\mathbf{m}}_{\mathcal{P}_j} \in \mathbb{R}^{k_j}$, the fitted latent predictors are

$$\widetilde{\boldsymbol{\eta}}_{i,\mathcal{P}_j} = \overline{\mathbf{m}}_{\mathcal{P}_j} + \widehat{\mathbf{R}}_j \left(\mathbf{m}_{i,\mathcal{P}_j} - \overline{\mathbf{m}}_{\mathcal{P}_j} \right) \in \mathbb{R}^{k_j}.$$

Let $\widetilde{\mathbf{X}}_j \in \mathbb{R}^{n \times k_j}$ contain the centered fitted predictors. Using the same dependent variable $\mathbf{d}_j$ as above, the scaling James–Stein (JS) estimator of the loadings or latent regression slopes in equation j is

$$\widehat{\mathbf{a}}_{j,\mathrm{JS}} = (\widetilde{\mathbf{X}}_j^\top \widetilde{\mathbf{X}}_j)^{-1} \widetilde{\mathbf{X}}_j^\top \mathbf{d}_j.$$

Aggregated James–Stein Estimation

The aggregated James–Stein (JSA) estimator replaces each scaling indicator with a weighted aggregate of indicators satisfying the proxy conditions above. For latent predictor η_k in equation j, let $\mathbf{y}_{i,kj} \in \mathbb{R}^{r_{kj}}$ collect these indicators, excluding the dependent variable when estimating a measurement equation. Let $\boldsymbol{\nu}_{kj}$, $\boldsymbol{\lambda}_{kj}$, and $\mathbf{w}_{kj}$, all in $\mathbb{R}^{r_{kj}}$, contain their intercepts, loadings, and weights. Normalizing the aggregate to the latent variable's scale gives

$$g_{ikj} = \frac{\mathbf{w}_{kj}^\top (\mathbf{y}_{i,kj} - \boldsymbol{\nu}_{kj})}{\mathbf{w}_{kj}^\top \boldsymbol{\lambda}_{kj}} = \eta_{ik} + \epsilon_{ikj}^{(A)}, \qquad \mathbf{w}_{kj}^\top \boldsymbol{\lambda}_{kj} \neq 0.$$

The weights are chosen to maximize reliability (Bentler, 1968). In the sample,

$$\widehat{\mathbf{w}}_{kj} \in \arg\max_{\mathbf{w} \neq \mathbf{0}} \left\{ 1 - \frac{\mathbf{w}^\top \widehat{\boldsymbol{\Theta}}_{kj}^{\mathrm{pre}} \mathbf{w}}{\mathbf{w}^\top \mathbf{S}_{kj} \mathbf{w}} \right\},$$

where $\mathbf{S}_{kj} \in \mathbb{S}_{++}^{r_{kj}}$ is the sample covariance matrix of the indicators in the aggregate and $\widehat{\boldsymbol{\Theta}}_{kj}^{\mathrm{pre}} \in \mathbb{S}_{+}^{r_{kj}}$ is the diagonal matrix of their preliminary measurement error variance estimates. Preliminary loading estimates $\widehat{\boldsymbol{\lambda}}_{kj}^{\mathrm{pre}} \in \mathbb{R}^{r_{kj}}$ from the scaling James–Stein estimator supply the normalization.

Collect the aggregates and their measurement errors in

$$\mathbf{g}_{ij} = (g_{ikj})_{k \in \mathcal{P}_j} \in \mathbb{R}^{k_j}, \qquad \boldsymbol{\epsilon}_{ij}^{(A)} = (\epsilon_{ikj}^{(A)})_{k \in \mathcal{P}_j} \in \mathbb{R}^{k_j}.$$

Let

$$\boldsymbol{\Sigma}_{gg,j} = \mathrm{Var}(\mathbf{g}_{ij}) \in \mathbb{S}_{++}^{k_j}, \qquad \boldsymbol{\Omega}_{A,j} = \mathrm{Var}(\boldsymbol{\epsilon}_{ij}^{(A)}) \in \mathbb{S}_{+}^{k_j}.$$

The corresponding conditional expectation expression is

$$E(\boldsymbol{\eta}_{i,\mathcal{P}_j} \mid \mathbf{g}_{ij}) = \boldsymbol{\mu}_{\eta,\mathcal{P}_j} + \mathbf{R}_{A,j}(\mathbf{g}_{ij} - \boldsymbol{\mu}_{\eta,\mathcal{P}_j}), \qquad \mathbf{R}_{A,j} = (\boldsymbol{\Sigma}_{gg,j} - \boldsymbol{\Omega}_{A,j}) \boldsymbol{\Sigma}_{gg,j}^{-1}.$$

As in the scaling version, this is an exact conditional expectation under joint normality and otherwise a best linear predictor.

Moreover, because the loadings and latent regression slopes are estimated from centered observations, the empirical aggregates can be calculated without first estimating intercepts:

$$\widehat{g}_{ikj}^{c} = \frac{\widehat{\mathbf{w}}_{kj}^\top (\mathbf{y}_{i,kj} - \overline{\mathbf{y}}_{kj})}{\widehat{\mathbf{w}}_{kj}^\top \widehat{\boldsymbol{\lambda}}_{kj}^{\mathrm{pre}}}.$$

Let $\widehat{\mathbf{g}}_{ij}^{c} \in \mathbb{R}^{k_j}$ collect these centered aggregates, and let $\mathbf{S}_{gg,j} \in \mathbb{S}_{++}^{k_j}$ denote their sample covariance matrix. Under the proxy conditions, the aggregate error covariance estimate is the diagonal matrix

$$\widehat{\boldsymbol{\Omega}}_{A,j} = \operatorname{diag}\left[\left(\frac{\widehat{\mathbf{w}}_{kj}^{\top}\widehat{\boldsymbol{\Theta}}_{kj}^{\text{pre}}\widehat{\mathbf{w}}_{kj}}{\left(\widehat{\mathbf{w}}_{kj}^{\top}\widehat{\boldsymbol{\lambda}}_{kj}^{\text{pre}}\right)^2}\right)_{k\in\mathcal{P}_j}\right].$$

The estimated reliability adjustment and centered fitted predictors are

$$\widehat{\mathbf{R}}_{A,j} = (\mathbf{S}_{gg,j} - c_n\widehat{\boldsymbol{\Omega}}_{A,j})\mathbf{S}_{gg,j}^{-1}, \qquad \widetilde{\boldsymbol{\eta}}_{i,\mathcal{P}_j}^{c,A} = \widehat{\mathbf{R}}_{A,j}\widehat{\mathbf{g}}_{ij}^{c}.$$

Collecting the fitted predictors in $\widetilde{\mathbf{X}}_{A,j} \in \mathbb{R}^{n\times k_j}$ gives

$$\widehat{\mathbf{a}}_{j,\text{JSA}} = (\widetilde{\mathbf{X}}_{A,j}^{\top}\widetilde{\mathbf{X}}_{A,j})^{-1}\widetilde{\mathbf{X}}_{A,j}^{\top}\mathbf{d}_j.$$

The fitted predictor matrices $\widetilde{\mathbf{X}}_j$ and $\widetilde{\mathbf{X}}_{A,j}$ are assumed to have full column rank.

Estimation of Variance, Covariance, and Intercept Parameters

For each of the three estimators of the factor loadings and latent regression slopes, the remaining variance and covariance parameters are estimated by reweighted least squares (Maydeu-Olivares et al., 2026), holding these estimates fixed throughout the iterations. Let

$$\widehat{\mathbf{H}} = \widehat{\boldsymbol{\Lambda}}(\mathbf{I}_q - \widehat{\mathbf{B}})^{-1} \in \mathbb{R}^{p\times q},$$

where $\widehat{\boldsymbol{\Lambda}}$ and $\widehat{\mathbf{B}}$ are obtained using the selected estimator, MIIV, JS, or JSA, assuming that $\mathbf{I}_q - \widehat{\mathbf{B}}$ is nonsingular. Conditional on these estimates, the implied covariance matrix is

$$\boldsymbol{\Sigma}(\mathbf{v}) = \widehat{\mathbf{H}}\boldsymbol{\Psi}(\mathbf{v})\widehat{\mathbf{H}}^{\top} + \boldsymbol{\Theta}(\mathbf{v}),$$

where $\mathbf{v} \in \mathbb{R}^{d_v}$ collects the free variance and covariance parameters.

With fixed entries and linear equality restrictions incorporated into the parameterization,

$$\operatorname{vech}\{\boldsymbol{\Sigma}(\mathbf{v})\} = \widehat{\mathbf{c}} + \widehat{\mathbf{G}}\mathbf{v},$$

where $t = p(p+1)/2$, $\widehat{\mathbf{c}} \in \mathbb{R}^t$ is the fixed component under the chosen parameterization, and $\widehat{\mathbf{G}} \in \mathbb{R}^{t\times d_v}$ maps the free parameters to the implied covariance moments.

For a weight matrix $\widehat{\mathbf{W}} \in \mathbb{S}_{++}^t$, the weighted least squares estimate is

$$\widehat{\mathbf{v}} = \left(\widehat{\mathbf{G}}^{\top}\widehat{\mathbf{W}}\widehat{\mathbf{G}}\right)^{-1}\widehat{\mathbf{G}}^{\top}\widehat{\mathbf{W}}\left\{\operatorname{vech}(\mathbf{S}) - \widehat{\mathbf{c}}\right\},$$

assuming that $\widehat{\mathbf{G}}$ has full column rank.

Reweighted least squares updates the normal theory weight matrix (Browne, 1974) using the current implied covariance matrix:

$$\widehat{\mathbf{W}}^{(\tau)} = \frac{1}{2}\mathbf{D}_p^{\top}\left[\{\widehat{\boldsymbol{\Sigma}}^{(\tau)}\}^{-1} \otimes \{\widehat{\boldsymbol{\Sigma}}^{(\tau)}\}^{-1}\right]\mathbf{D}_p, \qquad \widehat{\boldsymbol{\Sigma}}^{(\tau)} = \boldsymbol{\Sigma}(\widehat{\mathbf{v}}^{(\tau)}).$$

Here, $\otimes$ denotes the Kronecker product, and $\mathbf{D}_p \in \mathbb{R}^{p^2\times t}$ is the duplication matrix satisfying $\operatorname{vec}(\mathbf{S}) = \mathbf{D}_p \operatorname{vech}(\mathbf{S})$. At iteration τ, substituting $\widehat{\mathbf{W}}^{(\tau)}$ into the weighted least squares expression gives $\widehat{\mathbf{v}}^{(\tau+1)}$. The updates are repeated until convergence, assuming that the implied covariance matrices used to construct the weights remain positive definite. The resulting $\widehat{\mathbf{v}}$ determines $\widehat{\boldsymbol{\Theta}}$ and $\widehat{\boldsymbol{\Psi}}$, which are not constrained to be positive semidefinite.

Finally, the intercepts are estimated from the sample means, using the previously estimated factor loadings and latent regression slopes. Under the scaling indicator parameterization, the mean structure is saturated when all nonscaling indicator intercepts and all structural intercepts are free. Let $\overline{\mathbf{m}} \in \mathbb{R}^q$ contain the sample scaling indicator means. Then

$$\widehat{\boldsymbol{\mu}}_\eta = \overline{\mathbf{m}}, \qquad \widehat{\boldsymbol{\nu}} = \overline{\mathbf{y}} - \widehat{\boldsymbol{\Lambda}}\overline{\mathbf{m}} \in \mathbb{R}^p, \qquad \widehat{\boldsymbol{\alpha}} = (\mathbf{I}_q - \widehat{\mathbf{B}})\overline{\mathbf{m}} \in \mathbb{R}^q.$$

These expressions preserve the fixed zero scaling indicator intercepts and reproduce the observed sample means. Additional mean structure restrictions require a corresponding constrained estimation step. Together, these stages estimate the free entries of $\boldsymbol{\Lambda}$, $\mathbf{B}$, $\boldsymbol{\Theta}$, $\boldsymbol{\Psi}$, $\boldsymbol{\nu}$, and $\boldsymbol{\alpha}$. Although the estimated factor loadings and latent regression slopes are held fixed during covariance estimation, their sampling uncertainty is propagated when calculating standard errors.

Analytical Standard Errors

For the MIIV estimates of the factor loadings and latent regression slopes, the conventional homoskedastic covariance estimator is

$$\widehat{\mathbf{V}}_{j,\text{MIIV}} = \widehat{\sigma}^2_{u_j}(\mathbf{X}_j^\top \mathbf{P}_j \mathbf{X}_j)^{-1} \in \mathbb{S}_+^{k_j},$$

where

$$\widehat{\sigma}^2_{u_j} = \frac{\widehat{\mathbf{u}}_j^\top \widehat{\mathbf{u}}_j}{n - k_j - 1}, \qquad \widehat{\mathbf{u}}_j = \mathbf{d}_j - \mathbf{X}_j \widehat{\mathbf{a}}_{j,\text{MIIV}},$$

where the intercept is absorbed by centering and $n > k_j + 1$. Here, $\sigma^2_{u_j}$ is the composite error variance of the L2O equation, rather than an individual measurement error or structural disturbance variance.

This equation specific expression concerns the factor loadings and latent regression slopes. For the variance and covariance parameters $\mathbf{v}$, the delta method is applied to the second stage. Let $\widehat{\mathbf{H}}_2 = (\widehat{\mathbf{G}}^\top \widehat{\mathbf{W}} \widehat{\mathbf{G}})^{-1} \widehat{\mathbf{G}}^\top \widehat{\mathbf{W}}$ be evaluated at the converged weights and treated as fixed, as in Bollen and Maydeu-Olivares (2007) and Jin et al. (2021), let $\widehat{\boldsymbol{\Delta}}_1$ be the Jacobian of $\text{vech}\{\boldsymbol{\Sigma}(\boldsymbol{\theta})\}$ with respect to the factor loadings and latent regression slopes, evaluated at the estimates, and let $\widehat{\mathbf{K}}$ be the Jacobian of their MIIV estimates with respect to $\text{vech}(\mathbf{S})$ (Fisher & Bollen, 2020). Under correct specification, the estimated covariance matrix is

$$\widehat{\mathbf{V}}_{\mathbf{v},\text{MIIV}} = \frac{1}{n} \widehat{\mathbf{H}}_2 (\mathbf{I}_t - \widehat{\boldsymbol{\Delta}}_1 \widehat{\mathbf{K}}) \widehat{\boldsymbol{\Gamma}}_S (\mathbf{I}_t - \widehat{\boldsymbol{\Delta}}_1 \widehat{\mathbf{K}})^\top \widehat{\mathbf{H}}_2^\top \in \mathbb{S}_+^{d_v},$$

where $\widehat{\boldsymbol{\Gamma}}_S \in \mathbb{S}_+^t$ estimates the asymptotic covariance matrix of $\sqrt{n}\,\text{vech}(\mathbf{S})$. The factor $\mathbf{I}_t - \widehat{\boldsymbol{\Delta}}_1 \widehat{\mathbf{K}}$ propagates the sampling uncertainty of the estimated factor loadings and latent regression slopes through the residual moments. The standard errors of the MIIV intercept estimates follow from the same argument applied to the mean structure step.

Burghgraeve et al. (2021) derived a variance expression for the latent regression slopes that accounts for the estimated reliability adjustment and used bootstrap standard errors in their evaluations. Here, analytical standard errors are obtained by differentiating the complete estimation procedure with respect to the sample moments. For $T \in \{\text{JS}, \text{JSA}\}$, write

$$\widehat{\boldsymbol{\theta}}_T = g_{n,T}(\mathbf{s}),$$

where $g_{n,T}$ maps the h sample moments to the d parameter estimates. This mapping includes the preliminary error variance estimates, the estimates of the factor loadings and latent regression slopes, the subsequent variance and covariance estimates, and the intercept estimates.

Suppose

$$\sqrt{n}\{\mathbf{s} - \boldsymbol{\sigma}(\boldsymbol{\theta})\} \xrightarrow{d} N_h(\mathbf{0}, \boldsymbol{\Gamma}), \qquad \boldsymbol{\Gamma} \in \mathbb{S}_+^h,$$

where $\boldsymbol{\Gamma}$ is the joint asymptotic covariance matrix of the sample means and covariances. Under the required moment and differentiability conditions, the delta method covariance estimator is

$$\widehat{\mathbf{V}}_T = \frac{1}{n}\widehat{\mathbf{J}}_T\widehat{\boldsymbol{\Gamma}}\widehat{\mathbf{J}}_T^\top \in \mathbb{S}_+^d, \qquad \widehat{\mathbf{J}}_T = \left.\frac{\partial g_{n,T}(\mathbf{s})}{\partial \mathbf{s}^\top}\right|_{\mathbf{s}=\mathbf{s}_{\text{obs}}} \in \mathbb{R}^{d\times h}.$$

The Jacobian is evaluated analytically by default by differentiating the estimation steps. For reweighted least squares, the derivative is obtained by implicit differentiation of the converged estimating equations. The Jacobian propagates uncertainty through the preliminary error variance estimates, reliability adjustments, estimates of the factor loadings and latent regression slopes, variance and covariance estimates, and intercept estimates. For JSA, it also accounts for the estimated aggregation weights and normalization loadings.

The matrix $\boldsymbol{\Gamma}$ can be estimated using normal theory or empirical moments through order four (Browne, 1984). Normal theory estimates may be evaluated at the sample covariance matrix or the fitted model covariance matrix. For parameter $r = 1, \ldots, d$, the analytical standard error is

$$\widehat{\text{SE}}(\widehat{\theta}_{T,r}) = \sqrt{[\widehat{\mathbf{V}}_T]_{rr}} \in [0, \infty).$$

For MIIV factor loadings and latent regression slopes, moment based covariance estimates are also available as $n^{-1}\widehat{\mathbf{K}}\widehat{\boldsymbol{\Gamma}}_S\widehat{\mathbf{K}}^\top$. Together with the second stage expression and the mean structure step above, these provide delta method standard errors calculated separately for each parameter block.

Simulation Design and Performance Evaluation

In this section we describe the design of the simulation study and the measures used to evaluate the standard errors.

Population Model and Estimands

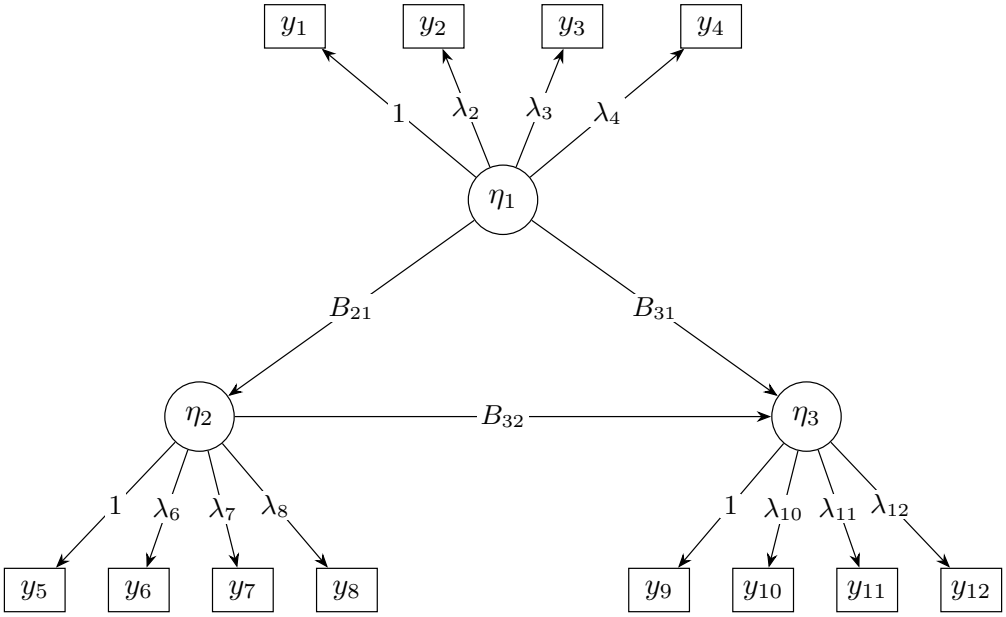
We study the model in Figure 1, with three latent variables and four indicators each. The latent variable model is

$$\begin{aligned} \eta_2 &= \alpha_2 + B_{21}\eta_1 + \zeta_2, \\ \eta_3 &= \alpha_3 + B_{31}\eta_1 + B_{32}\eta_2 + \zeta_3. \end{aligned} \tag{1}$$

The design is partly based on previous studies. Burghgraeve et al. (2021) used four indicators per latent variable, a common reliability of .6 or .8 for all indicators. Note also that the model is similar to the empirical illustration Burghgraeve et al. (2021). Moreover, Jin et al. (2021) used a recursive latent variable model with several equations and standardized slopes between 0.1 and 0.6, which the present model follows with $B_{21} = 0.6$, $B_{31} = 0.3$, and $B_{32} = 0.5$. With $\text{Var}(\eta_1) = 1$, $\text{Var}(\zeta_2) = 0.64$, and $\text{Var}(\zeta_3) = 0.48$, the variances of η_2 and η_3 equal 1, so that the slopes are standardized, and the implied correlations are .60 between η_1 and η_2, .60 between η_1 and η_3, and .68 between η_2 and η_3. The scaling indicators have loadings fixed to one and the remaining loadings range from 0.8 to 1.2. The mean structure has $E(\eta_1) = 1$, $\alpha_2 = \alpha_3 = 0.5$, and intercepts of 0.2 for the nonscaling indicators. The main estimands are all free factor loadings, latent regression slopes, residual variances, and latent variances and disturbance variances.

Figure 1

Path diagram of the simulation model with three latent variables and their indicators.



Note. Scaling indicator loadings are fixed to one. The population latent regression slopes are $B_{21} = 0.6$, $B_{31} = 0.3$, and $B_{32} = 0.5$.

Simulation Conditions and Estimation Approaches

We employed a fully factorial design with two factors: sample size ($n \in \{100, 200, 300, 400, 500, 600\}$) and indicator reliability ($\rho \in \{.6, .8\}$). This yielded $6 \times 2 = 12$ conditions, each replicated 1,500 times. The sample sizes cover the small to moderate range in which the equation by equation estimators are expected to differ most from maximum likelihood. The reliability levels match the low and high conditions of Burghgraeve et al. (2021) and determine the size of the reliability adjustment in the James–Stein estimators.

As highlighted, we investigated the JS, JSA, MIIV-2SLS, and ML approaches, all with the default options in lavaan.

Data Generation

Data were generated from the population model with all stochastic components normal and mutually independent. The exogenous latent variable was generated as $\eta_1 \sim \mathcal{N}(1, 1)$ and the structural disturbances as $\zeta_2 \sim \mathcal{N}(0, 0.64)$ and $\zeta_3 \sim \mathcal{N}(0, 0.48)$, from which η_2 and η_3 follow by Equation 1. The measurement error of each indicator was generated as normal with mean zero and variance $\lambda^2(1 - \rho)/\rho$, where λ is the loading of that indicator, so that every indicator has reliability ρ in its condition, and the indicators follow by the measurement equations. The observed variables are therefore multivariate normal with the mean vector and covariance matrix implied by the population values. All four estimators were applied to the same data sets.

Table 1

Performance Measures for Evaluating the Standard Errors

Performance measure	Empirical calculation	MCSE
SD	$S_{\hat{\theta}} = \sqrt{\frac{1}{K-1}\sum_{k=1}^{K}(\hat{\theta}_k - \bar{\theta})^2}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(S_{(j)} - S_{\hat{\theta}})^2}$
SE/SD ratio	$\overline{SE}/S_{\hat{\theta}}$	—
SE^2/VAR ratio	$\nu = \overline{SE^2}/S^2_{\hat{\theta}}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\nu_{(j)} - \nu)^2}$
Coverage	$c = \frac{1}{K}\sum_{k=1}^{K} I(L_k \leq \theta \leq U_k)$	$\sqrt{c(1-c)/K}$

Note. $\hat{\theta}_k$ is the estimate of parameter θ in replication k ($k = 1, \ldots, K$) and $\bar{\theta}$ the mean of the K estimates. $\widehat{SE}_k$ is the estimated standard error in replication k, $\overline{SE} = \frac{1}{K}\sum_k \widehat{SE}_k$, $\overline{SE^2} = \frac{1}{K}\sum_k \widehat{SE}_k^2$, and L_k, U_k are the confidence interval limits. The subscript (j) marks a leave-one-out jackknife replicate. The dash marks an MCSE that is not computed.

Convergence, Outliers, and Inadmissible Solutions

Replications were screened in two steps, and a replication that failed either step for any estimator was removed for all four. A solution was nonconverged when any estimate, standard error, or confidence interval bound of an estimand was missing or non-finite. Among the converged replications, outliers were point estimates more than three interquartile ranges below the first or above the third quartile, per condition, estimator, and estimand. Inadmissible solutions, with a nonpositive variance estimate, were retained and counted.

Performance Measures

Performance was assessed with the measures implemented in the simhelpers package (Joshi & Pustejovsky, 2025), computed for every estimand. These measures, with their empirical and Monte Carlo standard error (MCSE) calculations, are listed in Table 1. The MCSEs of the SD and of the SE^2/VAR ratio are obtained by jackknife and that of coverage in closed form. The SE/SD ratio and the SE^2/VAR ratio compare the same quantities on the standard deviation and on the variance scale, so only the latter, for which an MCSE is available, is shown in the figures.

Reproducibility and Computational Details

All analyses were conducted in R (R Core Team, 2025) using the lavaan package (Rosseel, 2012) for estimation, simhelpers (Joshi & Pustejovsky, 2025) for performance evaluation, and ggplot2 (Wickham, 2016) for the figures. The computational environment was pinned with Nix (Dolstra et al., 2004) via the rix package (Rodrigues & Baumann, 2026), and the article was written with the apaquarto extension (Schneider, 2026). The replication level estimates, standard errors, and confidence intervals, the performance measures, and the screening counts are saved as R data files in the project repository, together with the R version, package versions, code, and manuscript.

Results

Convergence, Outliers, and Inadmissible Solutions

All four estimators converged in every replication of every condition. Outlier screening removed 47 of the 18,000 replications (0.26%), concentrated at $n = 100$ with reliability .6, where 1.9% of the replications were removed, flagged mainly through JS (1.7% of the replications in that condition). In every other condition fewer than 0.5% of the replications were removed. Inadmissible solutions occurred in 4 replications, all for JS at $n = 100$ with reliability .6, where a residual or disturbance variance estimate was nonpositive. They are included in the results below.

Standard Error Performance

Each figure shows reliability .6 in the upper half and reliability .8 in the lower half. Figure 2 shows the latent regression slopes, Figure 3 one factor loading per latent variable, Figure 4 the residual variances of the scaling indicators and of one nonscaling indicator per latent variable, and Figure 5 the variance of η_1 and the two disturbance variances. The loadings and residual variances not shown behave like those shown for the same latent variable, with SE^2/VAR ratios within 0.14 and coverage within .03 of each other in every condition.

From $n = 300$ onward, the four estimators are indistinguishable within Monte Carlo error. The SE^2/VAR ratio stays within 0.11 of one and coverage within .035 of .95 for every estimator and parameter at both reliability levels, and the curves move together across sample sizes because they are computed on the same data sets. Three differences appear at $n = 100$ and $n = 200$, mainly at reliability .6.

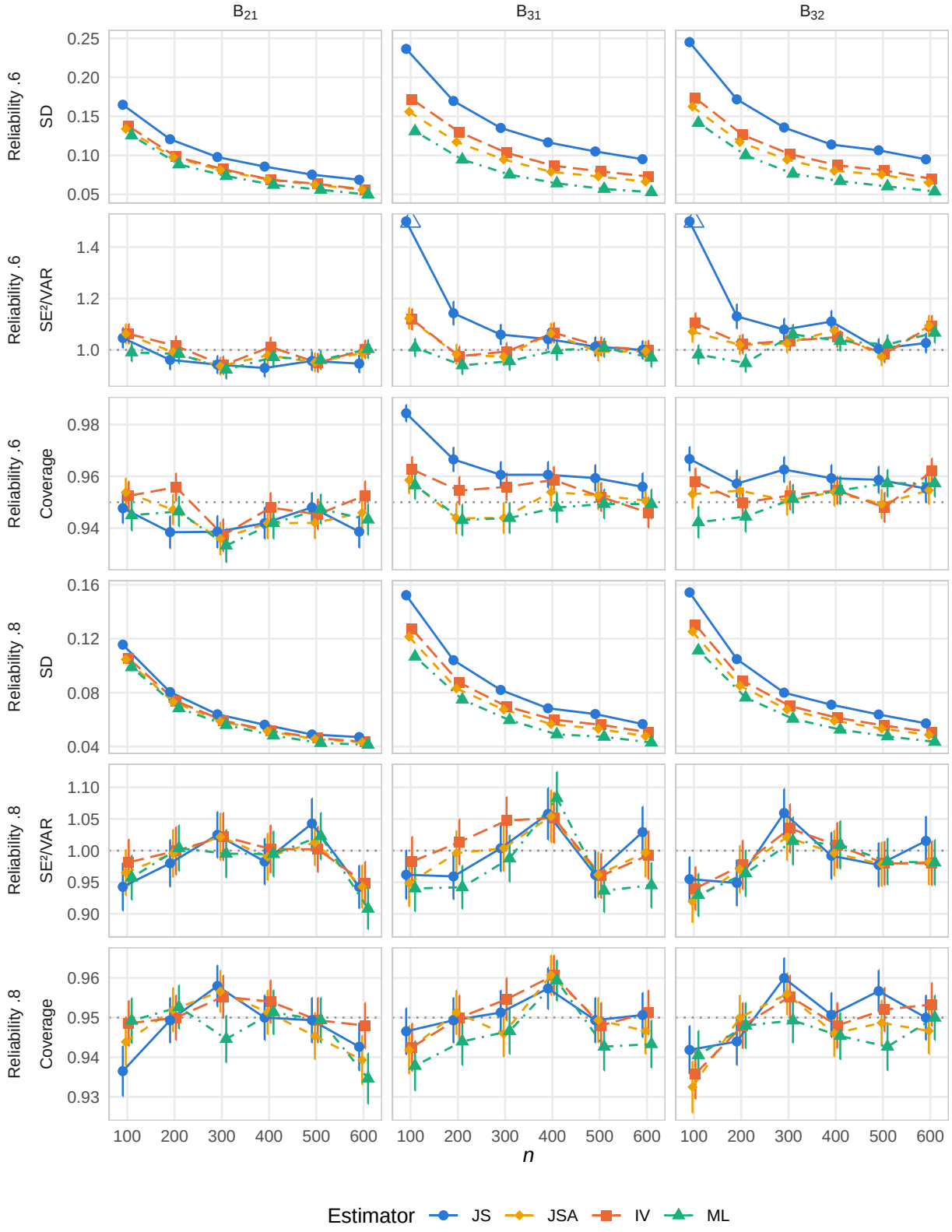
First, the scaling James–Stein standard errors for the slopes B_{31} and B_{32} are too large at reliability .6, with SE^2/VAR ratios of 1.5 to 1.6 at $n = 100$ and 1.13 at $n = 200$ and coverage of .96 to .98, while the SD is also the largest of the four, 0.24 against 0.13 to 0.17 at $n = 100$. In some replications the reliability adjustment for the equation of η_3 implies a correlation between η_1 and η_2 close to one, so that the fitted predictors are nearly collinear and both the estimates and their standard errors are inflated. The screening removed the most extreme of these replications, the milder ones remain. The aggregated estimator, which pools all indicators of a latent variable, is unaffected.

Second, the MIIV loadings undercover at reliability .6, with coverage of .88 to .89 at $n = 100$ against .93 to .95 for the other estimators, while the MIIV SE^2/VAR ratio is 1.05 to 1.07 and its SD the smallest. The gap comes from a downward bias of about 0.06 in the two-stage least squares loadings when the scaling indicators are unreliable, a weak instrument effect also reported by Jin et al. (2021), and it fades slowly with sample size.

Third, the disturbance variances separate the estimators at $n = 100$. ML undercovers them, at .91 to .94, although its SE^2/VAR ratio is close to one, MIIV overcovers them, at .96 to .97, with ratios of 1.11 to 1.18, and the James–Stein estimators lie in between. At reliability .8 the disturbance variance of η_2 is undercovered at .92 by ML and by both James–Stein estimators alike. The residual variances show no such contrast, with coverage between .92 and .96 for all estimators and a common dip at $n = 100$.

Figure 2

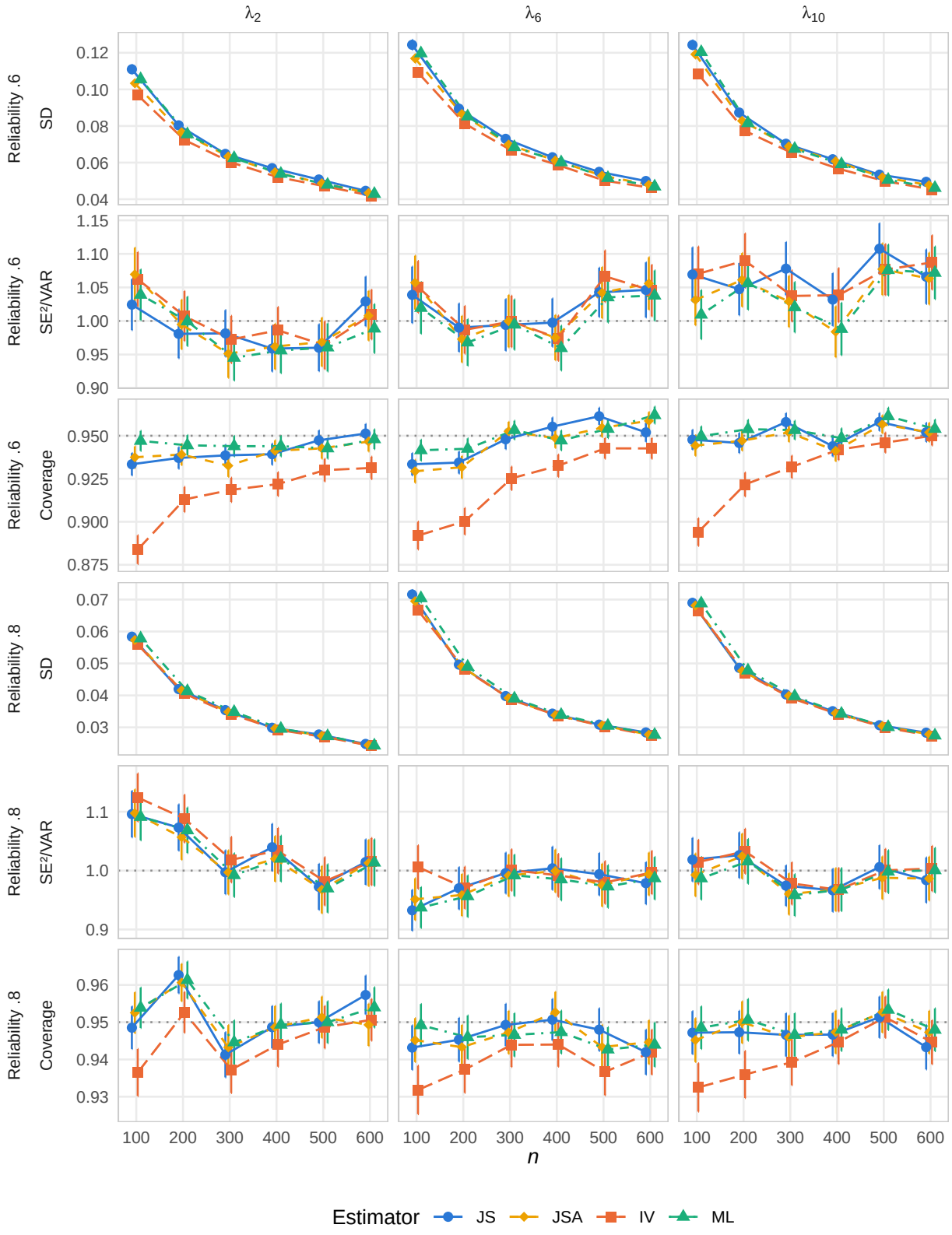
Performance of the Standard Errors of the Latent Regression Slopes



Note. JS = scaling James–Stein, JSA = aggregated James–Stein, IV = MIIV-2SLS, ML = maximum likelihood. Dotted lines mark the reference values 1 and .95. Error bars represent Monte Carlo standard errors. Values above 1.5 are drawn at 1.5 with an open triangle.

Figure 3

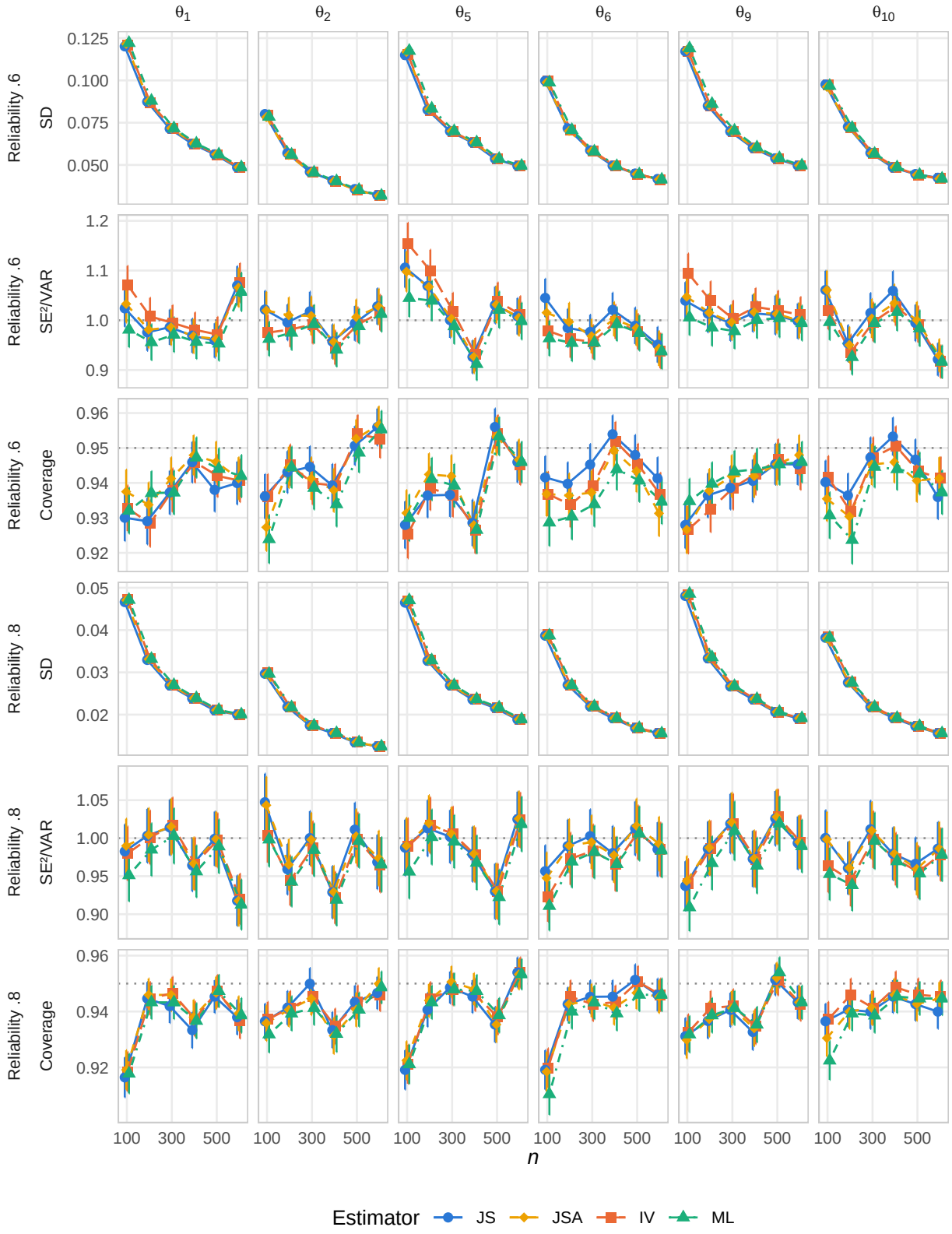
Performance of the Standard Errors of One Factor Loading per Latent Variable



Note. JS = scaling James–Stein, JSA = aggregated James–Stein, IV = MIIV-2SLS, ML = maximum likelihood. Dotted lines mark the reference values 1 and .95. Error bars represent Monte Carlo standard errors. Values above 1.5 are drawn at 1.5 with an open triangle.

Figure 4

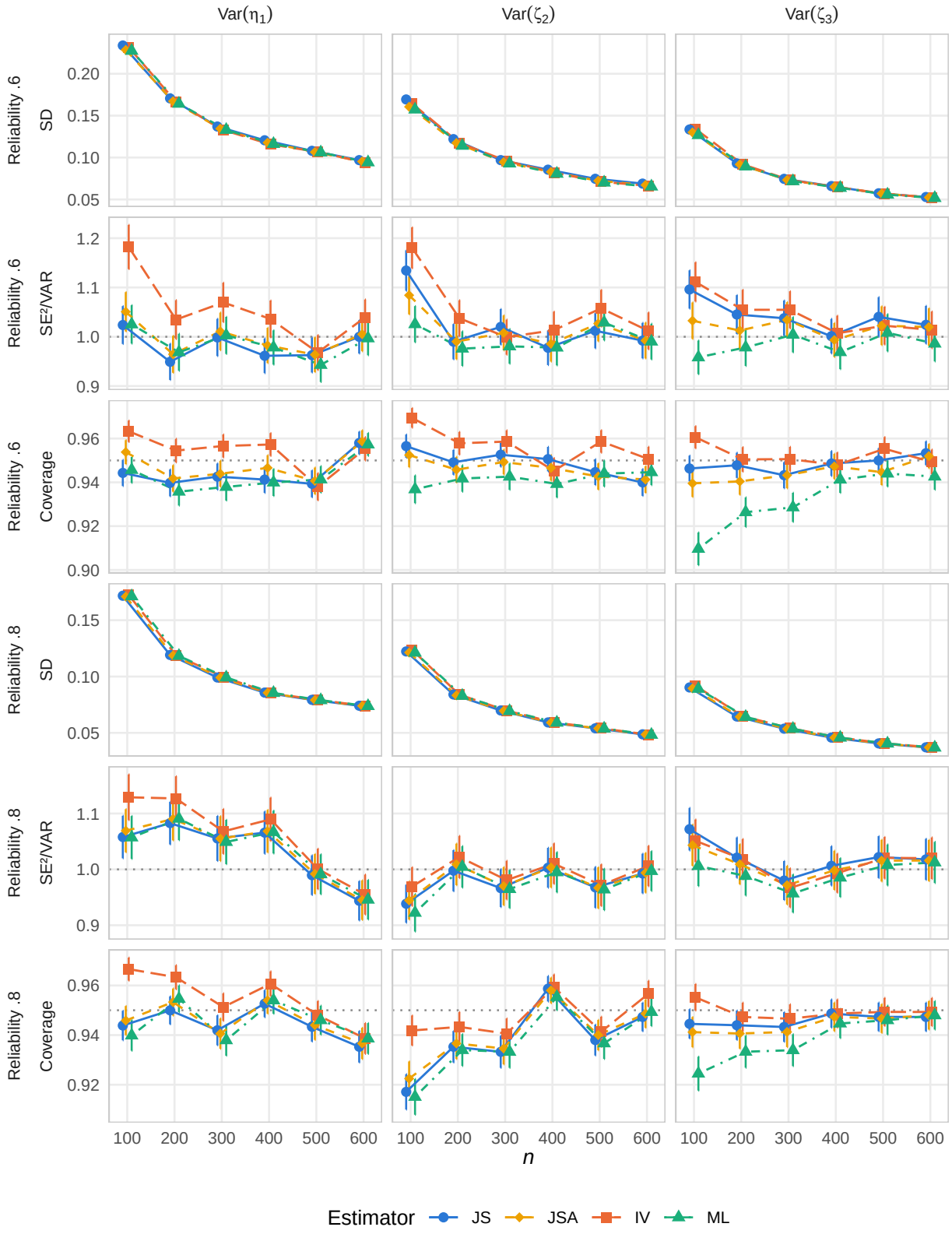
Performance of the Standard Errors of the Residual Variances of the Scaling Indicator and One Nonscaling Indicator per Latent Variable



Note. JS = scaling James–Stein, JSA = aggregated James–Stein, IV = MIIV-2SLS, ML = maximum likelihood. Dotted lines mark the reference values 1 and .95. Error bars represent Monte Carlo standard errors. Values above 1.5 are drawn at 1.5 with an open triangle.

Figure 5

Performance of the Standard Errors of the Latent and Disturbance Variances



Note. JS = scaling James–Stein, JSA = aggregated James–Stein, IV = MIIV-2SLS, ML = maximum likelihood. Dotted lines mark the reference values 1 and .95. Error bars represent Monte Carlo standard errors. Values above 1.5 are drawn at 1.5 with an open triangle.

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